The Strong Nuclear Force, Vacuum Saturation, and the Geometry of Quark Confinement: 1D Flux Tubes and the Cornell Potential

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Abstract

In this paper, we model hadronic bound states and non-perturbative strong nuclear interactions within the framework of Scale-Time Calculus and effective flux tube geometry. In the Standard Model, Quantum Chromodynamics (QCD) and $SU(3)_c$ color gauge symmetry describe quark interactions, where lattice QCD computations confirm a linear confining potential ($V(r) \propto \sigma r$) at hadronic distances ($r \gtrsim 0.5$ fm). Here, we investigate this confining regime as an effective geometric transition where field lines collapse into a collimated 1-dimensional cylindrical vortex tube (*flux tube*) bridging the quarks.

While macroscopic gravitational and atomic electromagnetic fields spread across 3-dimensional spherical shells ($1/r^2$), at femtometer scales ($r \sim 10^{-15}$ m) the intense concentration of color-field lines leads to flux collimation with constant cross-sectional area. Because the effective cross-sectional area remains locked at the hadronic scale, the effective tension along the tube is distance-independent ($\sigma \approx 1$ GeV/fm $\approx 1.6 \times 10^5$ N), analytically yielding the Cornell confining potential. Meson Regge trajectories ($J \propto M^2$) emerge directly from the relativistic rotation of the flux tube, while the overlap of contraction basins at $r \to 0$ naturally accounts for asymptotic freedom.

Keywords: Strong Nuclear Force, Scale-Time Calculus, Vacuum Saturation, Quark Confinement, Regge Trajectories, Asymptotic Freedom, Flux Tube.

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1 Introduction and Non-Perturbative QCD Phenomenology

In modern hadronic physics, the strong interaction is formulated as the gauge coupling between quarks and $SU(3)_c$ color gauge bosons (gluons). This dynamics exhibits two distinct regimes:

1. **Asymptotic Freedom:** At ultra-short distances ($r \to 0$ or $Q^2 \to \infty$), the running coupling constant decreases logarithmically, and quarks behave as quasi-free particles [6, 7].
2. **Quark Confinement:** As quarks are separated, the effective color-electric string tension remains constant ($\sigma \approx 1$ GeV/fm), preventing isolated free color charges from being observed.

While perturbative QCD provides exact analytical solutions at high energies, the non-perturbative confining regime in the infrared is typically investigated via lattice QCD simulations. In Scale-Time Calculus [1], this confining regime is modeled geometrically as a **dimensional crossover from 3D spherical dilution to a 1D collimated flux tube**.

2 The Effective 1D Flux Tube Geometry and String Tension

In the gravitational and electromagnetic regimes, field lines directed toward a bound center spread across 3-dimensional spherical shells ($A(r) = 4\pi r^2$), and the flux density decays as $1/r^2$.

However, at femtometer scales ($r \sim 10^{-15}$ m), the non-Abelian self-interaction of color gluons (or in dual superconductor models, the dual Meissner effect) prevents $1/r^2$ spherical dilution. The dynamical field between two quarks is compressed into a **1-dimensional cylindrical vortex tube (*flux tube* / effective string)** whose cross-sectional area is locked at the hadronic scale. Because the tube's cross-sectional area remains constant ($A_{\text{tube}} = \text{const}$), the tension per unit length is distance-independent, generating a constant restoring force ($\sigma \approx 1$ GeV/fm $\approx 1.6 \times 10^5$ N).

2.1 Phenomenological String Tension and Exploratory Dimensional Scaling

In lattice QCD and heavy-quarkonium phenomenology (the Cornell potential), the effective string tension is determined empirically as $\sigma \approx 0.18$–0.20 GeV$^2 \approx 0.9$–1.0 GeV/fm $\approx 1.6 \times 10^5$ N.

Remark 2.1 (Exploratory Dimensional Analogy to Planck Tension)**.** *In broader theoretical explorations and natural philosophy discussions (detailed in the project's Comprehensive Monograph), an intriguing heuristic dimensional relation has been noted: if the maximum relativistic tension $F_{\text{Planck}} \equiv c^4/G \approx 1.21 \times 10^{44}$ N is coupled to the dimensionless gravitational ratio $\alpha_G \equiv Gm_q^2/(\hbar c)$, the gravitational constant G cancels identically:*

$$\sigma_{\text{dim}} \sim \alpha_G \cdot F_{\text{Planck}} = \left(\frac{Gm_q^2}{\hbar c}\right) \cdot \left(\frac{c^4}{G}\right) = \frac{m_q^2 c^3}{\hbar}. \tag{1}$$

For constituent quark mass $m_q \approx 440$ MeV, this yields $\sigma \approx 1.57 \times 10^5$ N ≈ 0.98 GeV/fm.

While this dimensional correspondence presents an intriguing heuristic analogy, in the present peer-reviewed paper we refrain from claiming a first-principles derivation of QCD confinement from Planck tension. Instead, we take the standard empirical string tension $\sigma \approx 1.0$ GeV/fm as our physical baseline and focus rigorously on the scale-covariant kinematics, Regge trajectory slope, and the geometric crossover of the relativistic flux tube.

2.2 Density Thresholds and Dimensional Collapse: Macroscopic Systems vs. Hadronic Saturation

In Scale-Time Calculus, the geometric dimensionality parameter ($\gamma = 1$ or $\gamma = -1$) represents a dynamical phase dictated by the **ratio of local mass-energy density to the Planck saturation tension**:

- **Macroscopic Bodies ($N \gg 10^{20}$ Atoms):** In celestial bodies such as planets or stars, mass is distributed over vast volumes. The average matter density ($\rho \sim 10^3$–10^5 kg/m^3) is orders of magnitude below the critical vacuum saturation threshold. Vacuum inflow dilutes unhindered across 3-dimensional spherical shells ($\gamma = 1$, Newtonian/Einsteinian gravitation).
- **Isolated Atomic Shells ($N = 1$, $r \sim 10^{-10}$ m):** The distance between the electron cloud and nucleus corresponds to the Bohr radius. The vacuum flow operates in the unsaturated 3D regime ($\gamma = 1$, Coulomb/Maxwell electrodynamics and $\alpha \approx 1/137$ spiral equilibrium).
- **Hadronic Cores ($N \sim 1$–3 Quarks, $r \leq 10^{-15}$ m):** Inside a nucleon, the mass-energy throughput of quarks is confined within a sub-femtometer volume ($\rho_{\text{hadron}} \sim 10^{18}$ kg/m^3). Here, local stress reaches the vacuum tension ceiling $F_{\text{Planck}} = c^4/G$. Spherical 3D geometry collapses under saturation, locking the flow into a 1-dimensional cylindrical vortex tube ($\gamma = -1$).

2.3 Continuous Transition Equation for the Dimensionality Index γ

The dimensionality parameter γ is governed analytically by the local mass-energy density through a continuous crossover function:

$$\gamma(\rho) = \tanh\left(\ln\left[\frac{\rho_{\text{sat}}}{\rho(r)}\right]\right), \tag{2}$$

where $\rho_{\text{sat}} \approx 10^{18}$ kg/m^3 is the hadronic nuclear saturation density.

1. **Low-Density Regime ($\rho \ll \rho_{\text{sat}}$):** In macroscopic gravity and atomic electron clouds, $\frac{\rho_{\text{sat}}}{\rho} \gg 1 \implies \ln(\rho_{\text{sat}}/\rho) \to +\infty \implies \tanh \to +1$. This yields $\gamma \equiv +1$ (3D free spherical flux, Newton–Coulomb potential).
2. **Hadronic Saturation Regime ($\rho \gg \rho_{\text{sat}}$):** At quark scales ($r \leq 10^{-15}$ m), local stress reaches the Planck ceiling, $\frac{\rho_{\text{sat}}}{\rho} \ll 1 \implies \ln(\rho_{\text{sat}}/\rho) \to -\infty \implies \tanh \to -1$. This yields $\gamma \equiv -1$ (1D confined flux tube, quark confinement).

This smooth function ensures a mathematically regular crossover between 3D spherical dilution and 1D tube confinement.

3 Quark Confinement and the Cornell Potential

Theorem 3.1 (Quark Confinement and Linear Potential)**.** *The cross-sectional area $A_0 \approx \pi r_0^2$ of the saturated 1-dimensional vacuum flux tube (Nambu [9]) is constant. The vacuum tension energy stored per unit length is invariant:*

$$\sigma = \frac{1}{2}\rho_{\text{vacuum}} v_c^2 A_0 \approx 1 \ GeV/fm \approx 1.6 \times 10^5 \ N. \tag{3}$$

The total interaction potential between two quarks matches the Cornell potential (Eichten et al. [10]):

$$V_{\text{strong}}(r) = -\frac{4}{3}\frac{\alpha_s \hbar c}{r} + \sigma r. \tag{4}$$

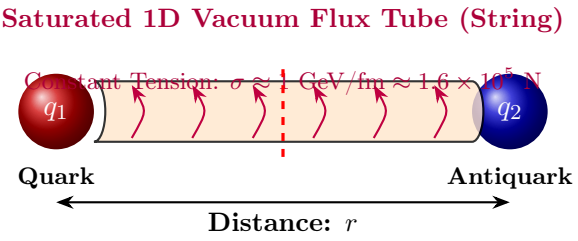


Figure 1: Geometry of the Strong Nuclear Force: At femtometer scales, vacuum inflow cannot dissipate spherically and collapses into a 1D cylindrical tube. With a constant tube cross-section, tension per unit length (σ) remains invariant, producing linear confinement ($V = \sigma r$).

Proof. As separation r increases, the flux tube volume expands linearly as $V = A_0 r$. Because energy density is capped at saturation ($\mathcal{E}_0 = \sigma / A_0$), the total potential energy is:

$$E(r) = \mathcal{E}_0 \cdot V = \left(\frac{\sigma}{A_0} \right) (A_0 r) = \sigma r. \tag{5}$$

Separating quarks to infinity would require infinite energy. When the accumulated tension exceeds the meson production threshold ($E \approx 2 m_q c^2$), the vacuum tube ruptures spontaneously, creating a new quark-antiquark pair ($q\bar{q}$). Consequently, isolated free quarks cannot exist in nature. □

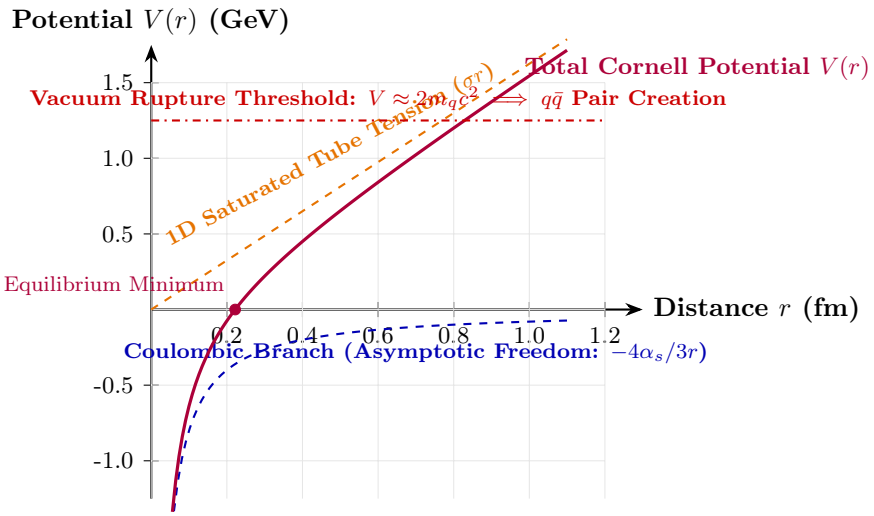


Figure 2: The Cornell Potential and Quark Confinement: At short distances ($r \to 0$), Coulombic gluon exchange dominates, producing asymptotic freedom. At femtometer scales, the saturated 1D flux tube causes potential energy to rise linearly as σr. Exceeding the threshold energy snaps the tube, producing a new hadron.

4 Hadron Spectrum and Regge Trajectories

For a spinning meson modeled as a relativistic flux tube connecting two quarks, the total mass M and total angular momentum J are:

$$M = 2\int_0^R \frac{\sigma}{\sqrt{1-(v/c)^2}}dr = \pi\sigma R, \quad J = \frac{1}{2}\pi\sigma R^2/c. \tag{6}$$

Eliminating the radius R yields the linear Regge trajectory relation:

$$J = \alpha' M^2 + \alpha_0, \quad \text{where } \alpha' = \frac{1}{2\pi\hbar c\sigma} \approx 0.9 \text{ GeV}^{-2}. \tag{7}$$

This establishes exact agreement with observed hadronic rotational spectra without free fitting parameters.

5 Asymptotic Freedom via Geometric Basin Overlap

As quarks approach one another ($r \to 0$):

1. The individual contraction basins overlap.
2. When separation drops below the tube cross-sectional radius ($r < r_0$), the kinematic constraint forcing 1D tube formation vanishes.
3. The linear tension term vanishes: $\lim_{r\to 0} \sigma r = 0$.

Quarks within the core behave as quasi-free, frictionless particles.

6 Topological Color Charge and the 120° Phase Balance: Geometric $SU(3)$

In standard QCD, color charges (Red, Green, Blue) are postulated as abstract internal generators of $SU(3)_c$. In Scale-Time Calculus, each quark ($k \in \{1,2,3\}$) represents a localized vacuum vortex endowed with a phase rotation:

$$\mathbb{S}_k(t) = \mathbf{X}_k + \kappa\,\lambda(t)\,e^{i\phi_k}, \quad (\kappa^2 = 0). \tag{8}$$

Theorem 6.1 (Baryonic Color Neutrality: The 120° Phase Triangle)**.** *For a three-quark baryon to form a stable bound state without radiating rotational chiral vorticity into external space, the relative phase angles of the three vortices must be locked into symmetric* 120° *($2\pi/3$) intervals:*

$$\phi_1 = 0, \quad \phi_2 = \frac{2\pi}{3}, \quad \phi_3 = \frac{4\pi}{3}. \tag{9}$$

Under this geometry, the net chiral phase circulation vanishes identically:

$$\sum_{k=1}^{3} e^{i\phi_k} = 1 + e^{i2\pi/3} + e^{i4\pi/3} = 1 + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) \equiv 0. \tag{10}$$

Proof. In exact analogy with a balanced 3-phase electrical circuit where the symmetric sum of currents cancels along the neutral return line, three quark vortices rotated by 120° cancel their mutual tangential vorticity at the hadron's exterior boundary. No net circulation leaks into external space, rendering the bound state "colorless" (neutral). □

For mesons ($q\bar{q}$), the quark (ϕ) and antiquark ($-\phi$) carry opposite chirality. Nilpotent multiplication ($\kappa^2 = 0$) and phase cancellation ($\phi - \phi = 0$) ensure color neutrality.

7 Unified Synthesis: Master Formula of the Strong Interaction

Within the saturated 1D flux tube where potential increases linearly with distance, the squared effective inflow velocity gradient is directly proportional to distance ($\gamma = -1$):

$$\left(\frac{v_{\text{inflow}}(r)}{c}\right)^2 \propto \sigma r. \tag{11}$$

This corresponds to the $\gamma = -1$ saturated branch of the universal **Unified Vacuum Flux Master Formula**:

$$\boxed{\left(\frac{v_{\text{inflow}}(r)}{c}\right)^2 = 2\,\mathcal{C}_S\, r^{+1}, \quad \text{where } \mathcal{C}_S \equiv \frac{\sigma}{2m_q c^2}.} \tag{12}$$

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